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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS - GRANGEGORMAN

TU856 - BSc. (Honours) Degree in Computer Science

Year 3

SEMESTER 1 EXAMINATIONS 2025/26

CMPU3038 - Software Engineering III

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Exam Duration: 2 hours

Instructions: e.g. Answer three out of four questions. All questions carry 33 marks each. One complimentary mark shall be awarded.

Special Instructions/Handouts:

If asked in any question to provide an example of code, you may use any appropriate language of your choice or pseudo code in your answer.

- Q.1 (a)** (i) Using examples of your choice, explain the difference between an *Association* and a *Dependency* in the context of an object-oriented structural model. Illustrate your examples using a class diagram. (7 marks)
- (ii) If a domain class contained a member variable whose *type* was that of another domain class, explain how the default relationship between those two classes is a unidirectional, optional, one-to-one association. (5 marks)
- (iii) Explain, using sample code, how a one-to-one mandatory association could be implemented. (5 marks)
- (b)** (i) Define the term *Interface Class*. (3 marks)
- (ii) Given the modelling requirement stated below, explain how *Java Database Connectivity (JDBC)* is an example of its implementation.
- "Several classes may be required to present the same interface".* (5 marks)
- (c)** (i) Outline the **four (4)** laws of the *Principle of Least Knowledge*. (5 marks)
- (ii) Briefly explain how the *Principle of Least Knowledge* supports the goal of developing robust software. (3 marks)

- Q.2 (a)** (i) A *Design Pattern* is a concept that realises a fundamental principle of software design – “*don't reinvent the wheel*”. Describe how a design pattern achieves this. In your answer, include an outline of **three (3)** basic concepts that form the basis of a design pattern. (9 marks)
- (ii) When initiating a design for a new, relatively complex system, describe an approach a software engineer could take which involves “*thinking in patterns*”. (4 marks)
- (b)** The *Data Access Object (DAO)* design pattern is generally considered a *structural* design pattern even though it exhibits some behavioural intent.
- (i) Explain the difference between *structural* and *behavioural* design patterns. (6 marks)
- (ii) Discuss what you consider to be the behavioural aspects of the *DAO* design pattern. (4 marks)
- (c)** State the *intent* of the *Bridge* design pattern and using a structural model (class diagram), explain how it can benefit a design. (10 marks)
- Q.3 (a)** The *Front Controller* is a well utilised *behavioural* design pattern.
- (i) Outline why it is categorised as *behavioural*. (3 marks)
- (ii) Other than the *Context Object* design pattern, provide the intent of **two (2)** design patterns that could collaborate with the *Front Controller* pattern in the context of a web application. (6 marks)
- (iii) Provide a sequence diagram that would communicate how an implementation of the **two (2)** patterns identified in *part (ii)* above, and the *Front Controller* pattern would collaborate. (10 marks)
- (iv) State the *intent* of the *Context Object* design pattern and describe how it could be used to remove a limitation that could arise when using the *Front Controller* pattern in a web application design. (6 marks)
- (b)** (i) Describe what is meant by a *RESTful* web service. (4 marks)
- (ii) *JAX-RS* is a Java API for supporting the development of *RESTful* web services. Briefly discuss how the *Front Controller* pattern is used in this implementation. (4 marks)

- Q.4 (a)**
- (i) *Object Relational Mapping (ORM)* is generally concerned with mapping object-oriented classes to relational database tables. State **four (4)** specific issues that it addresses.

(4 marks)
 - (ii) Using an example of your choice, describe how an inheritance hierarchy (a superclass and **two (2)** subclasses) that exists within a domain model could be mapped to a relational database where all classes are mapped to a separate table. Provide **two (2)** advantages and **two (2)** disadvantages to this approach.

(9 marks)
 - (iii) Describe an alternative approach to that given in *part (ii)* above, giving **one (1)** advantage and **one (1)** disadvantage of the approach.

(4 marks)
- (b)** Object-oriented software development (OOD) is one of several paradigms used to design and programme software systems. It has unique aspects that impact testing strategies.
- (i) The models produced as part of the analysis and design phases of OOD should be tested as they are developed. Discuss **why** and **how** these models should and can be tested.

(10 marks)
 - (ii) *Generalisation / Specialisation* relationships lead to specific circumstances when testing object-oriented systems. Using an example of your choice, discuss the main consideration that should be made in this context.

(6 marks)